

STATE FORENSIC TOXICOLOGY LABORATORY

TOXICOLOGY REPORT

Laboratory Case Number:	TOX-2024-11-3847
Medical Examiner Case:	ME-2024-001-ALK
Decedent:	CONFIDENTIAL
Age/Sex:	42-year-old Male
Date of Death:	November 13, 2024 (estimated)
Date Specimens Received:	November 16, 2024
Date Analysis Completed:	November 18, 2024
Toxicologist:	Dr. Amanda Foster, PhD, DABFT
Laboratory Director:	Dr. Robert Martinez, PhD, DABFT

SPECIMENS RECEIVED

The following postmortem specimens were received from the Medical Examiner's office on November 16, 2024, in properly sealed and labeled containers with intact chain of custody documentation:

- Femoral blood (30 mL, gray-top tube containing preservative and anticoagulant)
- Heart blood (20 mL, gray-top tube)
- Vitreous humor (5 mL from both eyes, combined)
- Urine (40 mL)
- Liver tissue (approximately 50 grams)
- Gastric contents (approximately 100 mL)
- Bile (10 mL)

All specimens were properly refrigerated upon receipt and maintained at 2-6°C until analysis. Chain of custody was maintained throughout the analytical process.

ANALYSES PERFORMED

Comprehensive toxicological screening was performed utilizing the following analytical methodologies:

- Headspace Gas Chromatography with Flame Ionization Detection (HS-GC-FID) for volatile alcohols
- Enzyme-Linked Immunosorbent Assay (ELISA) for drugs of abuse screening
- Gas Chromatography-Mass Spectrometry (GC-MS) for confirmation of drugs and volatile compounds
- Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS) for pharmaceutical compounds
- Spectrophotometric analysis for carbon monoxide
- Inductively Coupled Plasma Mass Spectrometry (ICP-MS) for heavy metals

TOXICOLOGY FINDINGS

Alcohol Analysis

Specimen	Ethanol Concentration	Method
Femoral Blood	0.42% (420 mg/dL)	HS-GC-FID
Heart Blood	0.38% (380 mg/dL)	HS-GC-FID
Vitreous Humor	0.38% (380 mg/dL)	HS-GC-FID
Urine	0.55% (550 mg/dL)	HS-GC-FID

Other Volatiles Tested: Methanol, Isopropanol, Acetone - All NEGATIVE

Drugs of Abuse Screening

Drug Class	Screening Result	Confirmation
Amphetamines	NEGATIVE	Not performed
Benzodiazepines	NEGATIVE	Not performed
Barbiturates	NEGATIVE	Not performed

Cannabinoids (THC)	NEGATIVE	Not performed
Cocaine/Metabolites	NEGATIVE	Not performed
Opioids/Opiates	NEGATIVE	Not performed
Phencyclidine (PCP)	NEGATIVE	Not performed
Methadone	NEGATIVE	Not performed
Synthetic Cannabinoids	NEGATIVE	Not performed

Pharmaceutical Screening (LC-MS/MS)

Comprehensive pharmaceutical screening for common prescription and over-the-counter medications was performed. The following classes were included in the analysis:

- Antidepressants (SSRIs, SNRIs, TCAs)
- Antipsychotics
- Cardiac medications (beta-blockers, calcium channel blockers, ACE inhibitors, digoxin)
- Anticonvulsants
- Muscle relaxants
- Antihistamines
- NSAIDs and analgesics
- Hypoglycemic agents

Result: NEGATIVE for all tested pharmaceutical compounds

Additional Testing

Test	Result	Method
Carbon Monoxide (COHb)	NEGATIVE (<5%)	Spectrophotometry
Cyanide	NEGATIVE	Spectrophotometry
Heavy Metals Panel	NEGATIVE	ICP-MS
(Lead, Arsenic, Mercury)		

INTERPRETATION AND DISCUSSION

Ethanol Findings

The femoral blood ethanol concentration of 0.42% (420 mg/dL) represents an extremely elevated level of acute alcohol intoxication. This concentration is more than five times the legal limit for driving (0.08% or 80 mg/dL) in most jurisdictions and falls well within the range associated with potentially lethal toxicity. At blood alcohol concentrations above 0.40%, individuals typically experience severe central nervous system and respiratory depression, loss of consciousness, and are at high risk for death.

The vitreous humor ethanol concentration of 0.38% correlates well with the blood findings. Vitreous humor is less subject to postmortem redistribution and contamination than blood, lending additional confidence to the results. The slightly lower concentration in vitreous compared to femoral blood is within expected variation and does not suggest postmortem production of alcohol or significant redistribution artifacts.

The elevated urine ethanol concentration (0.55%) compared to blood is consistent with active alcohol absorption and distribution at the time of death, though postmortem redistribution may also contribute. The heart blood concentration (0.38%) being slightly lower than femoral blood is not unusual and does not significantly impact interpretation.

At the documented blood alcohol concentration, expected physiological effects include:

- Profound CNS depression with loss of consciousness
- Severe respiratory depression and potential respiratory arrest
- Impaired protective airway reflexes increasing aspiration risk
- Cardiovascular depression
- Hypothermia
- Severe impairment of all cognitive and motor functions
- High risk of death without medical intervention

Absence of Other Substances

The comprehensive toxicological screening was entirely negative for drugs of abuse, prescription medications, over-the-counter drugs, and other toxic substances. This indicates that ethanol was the sole intoxicating substance present and that no other drugs or toxins contributed to death. The negative findings also exclude poisoning by pharmaceutical agents, carbon monoxide, cyanide, or heavy metals.

Tolerance Considerations

While chronic alcohol users may develop some degree of tolerance to alcohol's effects, allowing them to function at blood alcohol concentrations that would incapacitate non-tolerant individuals, this

tolerance is incomplete and does not confer protection against lethal toxicity at extremely high concentrations such as documented in this case. Even in individuals with significant tolerance, blood alcohol concentrations exceeding 0.40% carry substantial risk of death through respiratory depression and other mechanisms.

CONCLUSIONS

Toxicological analysis confirms the presence of ethanol at an extremely elevated and potentially lethal concentration in all specimens analyzed. The blood ethanol concentration of 0.42% is consistent with severe acute alcohol intoxication and is sufficient to cause death through central nervous system and respiratory depression. No other drugs, medications, or toxic substances were detected that would contribute to death or complicate interpretation.

These findings support the autopsy findings and are consistent with death due to acute ethanol intoxication.

Amanda Foster, PhD, DABFT

Forensic Toxicologist

Diplomate, American Board of Forensic Toxicology

Reviewed and Approved:

Robert Martinez, PhD, DABFT

Laboratory Director

Date: November 18, 2024